“Introduction of Apiculture”

Apiculture

- The word ‘apiculture’ comes from the Latin word ‘apis’ meaning bee.
- “Apiculture or bee keeping is the care and management of honey bees for the production of honey and the wax ’’.

Beehive is an enclosed, man-made structure in which some honey bee species live and raise their young.

Nests are cavities of hanging and exposed colonies of honey bees which house themselves naturally.

Apiary is an area where a large number of beehives can be placed. Here, the bees are taken care of and managed to produce wax and honey.

Beehive

Nests

Apiary

The main advantages of bee keeping

- They provide honey which is the most valuable nutritional food.
- They provide bee wax which is used in many industries, including cosmetics industries, polishing industries, pharmaceutical industries, etc.

- They Plays an excellent role in pollination.
- Honey bees are the best pollinating agents which help in increasing the yield of several crops.

- According to the recent studies, the honey bee's venom contains a mixture of proteins which can potentially be used as a prophylactic to destroying the HIV virus that causes AIDS in humans.

Classification of honey bees:

Kingdom:	Animalia
Phylum:	Arthropoda
Class:	Insecta
Order:	Hymenoptera
Family:	Apidae
Species:	Apis

Species of Honey bees:

- The beekeepers mostly take care of only those bee species whose names start with “**Apis**”- as they are the only species which produce honey.
- Common species of honey bees that are reared are as follows:

  Apis dorsata: Rock bee.

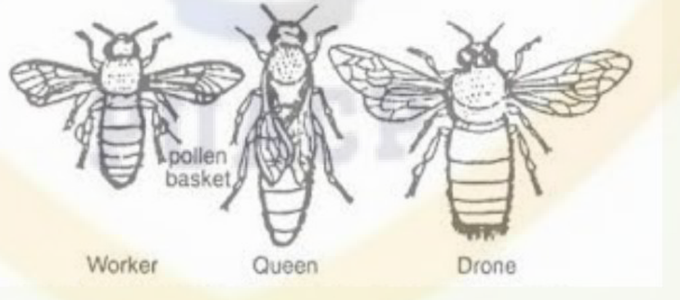
  Apis indica: Indian bee.

  Apis florea: Little bee.

  Apis mellifera: Italian bee.

Socialization of Honeybees

- **Queen** is 15-20 mm in length and consists of antennae, compound eyes, head, thorax long tapering abdomen, short legs and wings. It feeds on Royal Jelly and lays 1500 and 21000 eggs in a day.
- **Workers** have long proboscis for sucking nectar, strong wings, pollen-basket, sting and wax gland.
- **Drones** are male bees called as king which are without pollen collecting apparatus and wax gland. Function of Drone is to fertilize queen. Drone measures 15-17 mm in length.
- **Order of size:** Queen > Drones > Workers



The Queen Bee

- It is a diploid, fertile female.
- The presence of queen is a must in colony.
- The size of the body of queen is much larger than other castes of bees of the colony.
- The queen has a sting, curved like a sword at the tip of the abdomen, which is a modification of the egg-laying organ known as **ovipositor**. The sting serves as an organ of defence. She never uses it against anybody except her own caste.
- She lays both fertilized eggs (from which females develop) and unfertilized eggs (from which males develop).

The Worker Bee

- It is a diploid, sterile female.
- The size of a worker is the smallest among all the other castes but they constitute majority population of the bees in a colony.

Functions of worker bees

- Collection of honey
- Producing royal jelly for feeding the community
- Raising larvae and young ones
- Cleaning the comb
- Making wax
- Constructing the beehive
- Defending and protecting the hive
- Clearing the debris and dead bees
- Maintaining the temperature of the hive

Types of Worker Bee

- Laying workers
- Nurse workers
- House workers
- Field workers
- Soldier Workers

The Drone Bee

- It is haploid, fertile male.
- The males are larger than workers and are quite noisy.
- They have large wings, robust body and reduced mouthparts.
- They are unable to gather food, but they voraciously eat the food fed to them by the worker bees.
- They are stingless and their sole function is to fertilize the queen.
- The drone develops parthenogenetically from unfertilized eggs.
- Drones live only for a short period of time.

Flora of Apiculture

- The local surrounding flora is very important for selecting the location of the apiary.
- It is best practice to choose more nectar-yielding and pollen producing plants.
- Many forest-based and cultivated plants can provide nectar and pollen for foraging bees.
- Pakistan is also endowed with more than 700 plants species which are known to be visited by honeybees.
- This available bee flora can support 0.4 to 0.5 million honeybee colonies, producing 6,000-10,000 tons of honey per annum.

Products of Beekeeping

Following products are collected as a result of beekeeping

- Pollen
- Honey
- Royal Jelly
- Bee Wax
- Propolis
- Bees Venum

Honey

Royal Jelly

Propolis

Bees Venum

Pollen

- Bees collect 66 lbs of pollen per year, per hive.
- Honeybee uses pollen as a food which is called bee-bread.

Honey

- Honey bees swallow nectar and store it in their crops where secretions of stomach lining, salivary glands and mandibular glands are mixed in it.
- It is then regurgitated into the cell and fanned to evaporate excess moisture. This process is repeated many times to ripen the honey.

- **Royal Jelly**
- The powerful, milky substance that turns an ordinary bee into a Queen Bee.
- It is made of digested pollen and honey or nectar mixed with a chemical secreted from a gland in a nursing bee's head.
- **Bees Wax**
- It is secreted from the glands on abdominal segments 4-7 ü Bees wax is used by the honey bees to build honey comb.

Propolis

- It is collected by honeybees from trees, the sticky resin is mixed with wax to make a sticky glue that makes foundation of the comb strong.
- The bees use this to seal cracks and repair their hive.

Bees Venom

- Honey bee venom contains at least 18 active substances.
- It contains potent anti-inflammatory agents
- It has analgesic activity
- Enhances nerve transmission.

Life Cycle

Life-cycle stages comprises

- Egg
- Larva
- Pupa
- Adult

Some Terms to Know

- **Swarm** – A large group of worker bees, drones, and a queen that leaves the mother colony to establish a new colony.

- **Skeps** look like overturned pots and were made from either baked clay or woven straw.

- **Brood** – Young developing bees found in their cells in the egg, larval, and pupal stages of development.

- **Honey Comb-** A structure of hexagonal cells of wax, made by bees to store honey and eggs.

- **Beehive** is an enclosed, man-made structure in which some honey bee species live and raise their young.

- **Colony** – A community of honey bees having a queen, thousands of workers, and (during part of the year) a number of drones.

- **Super** – A receptacle in which bees store surplus honey placed “over” (above) the brood chamber. As a verb, to add supers in expectation of a honey flow.

- **Nests are** cavities of hanging and exposed colonies of honey bees which house themselves naturally.

History of Beekeeping

Traditional Beekeeping

- Based On known sources, today it is known that the first bee keepers were ancient Egyptians.. They were also first as nomadic beekeepers.
- They were moving massive amount of colonies between upper and lower Egypt, according to the season.
- They also learned how to harvest honey without destroying hives.
- Later Romans, as well as many other communities in Asia and Europe started keeping bees too.
- Modern form of beekeeping started in late 18th and beginning of 19th century.

- With the discovery of American and Australia, they exported honeybees to those continents as well.
- It was Portuguese that exported bees to Brazil.
- While to North America it was colonists that brought honeybees in first half of 17th Century.
- To Australia they were brought in 1822 and to New Zealand in 1842 .
- This way beekeeping was spread to most of Earth.
- In early medieval age men were using honey as a basic sweetener they produced candles.
- With spread of beekeeping, humanity also spread wonderful knowledge of health benefits of honeybee products.

Types of hives for bee keeping

Conventional Bees Hives:

- **Skeps** looks like overturned pots and were made from either baked clay or woven straw.
- A small hole near the bottom of the skep allowed the bees to come and go, and the comb is laid down inside.
- The use of a skep, unfortunately, still required the destruction of the hive, and often the death of the entire colony, to harvest the honey. So, beekeepers began looking for alternative hive designs.

Modern Bee Hives

- On 5th October 1852, the Reverend Lorenzo, an American Congregational minister, received a patent on the first movable frame beehive in America.
- A patent is anything which cannot be copied without permission. So, after his invention beekeeping industry was revolutionized and first time allow the beekeeper to manipulate and use every aspect of the life of bee colony.
- It permitted colonies to be easily transverted and the

- Wooden hive boxes became more common by the 18th century, leading to Francois Huber's moveable hive or "leaf hive," a vertical stack of moveable book-like leaves, each holding its own section of comb.
- These containing honey. And could be removed without destroying the colony.

- Moving into the 19th century, other innovative apiarists, including Thomas Wildman, began experimenting with " bar hives," wooden boxes that are equipped with a row of bars across the top under which the bees build their comb in small hanging sections.
- As a result each section could be removed by lifting up on the bar but not without any effort.

Methods of Beekeeping

Old or indigenous method

- This is primitive and unplanned method of apiculture. In this method two types of hives are used,

- **Natural**
- **Artificial**

- In the indigenous method, the bees are first killed or made to escape from the hive with the help of smoke when the bees are at rest during night.
- This method has many drawbacks and it is not suitable for commercial large-scale production of honey.

Disadvantages of Indigenous Methods

The following are the disadvantages of indigenous method:

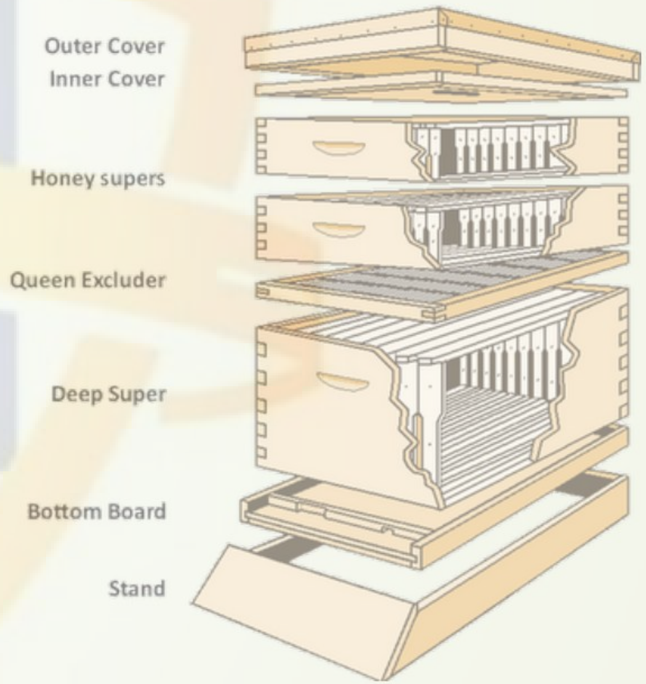
- The honey cannot be extracted in the pure form because it is contaminated with larvae, pupa and pollens.
- The future yield of the honey is affected as the colony has to be destroyed to extract the honey.
- Moreover it takes lot of energy of the bees to build new hive.
- The bees may not construct the new hive in the same place as the old one.

- The natural hives also have the danger of attack by the enemies like rats, monkeys, ants etc.
- The natural hives can also be damaged by the climatic factors such as wind flood.
- Also scientific intervention is difficult in the indigenous method and thus improving of the bee race is impossible. Because no new genes were introduced in colony if reared them again and again.

Though the indigenous method has many drawbacks, it still persists.

Modern method

- In the modern method of apiculture the honey bees are reared in movable artificial hives.
- The modern beehive is made up of a series of square boxes without tops or bottoms, set one above the other.
- These hives have a **floor board** and a **bottom stand** at the bottom and a **crown board** at the top.
- Inside these boxes, **frames** are vertically hung parallel to each other and can be shifted from one place to other easily.



- The frames are filled with sheets of wax foundation on which the combs are built by the bees.
- The entrance of the hive can be reduced with the help of the **entrance reducer**.
- The queen is usually confined to the brood chamber.
- The boxes termed **honey supers** are used for storage of honey.
- The queen is prevented from going to honey supers by the **queen excluder** that allows only the workers to move.

Inspecting a Colony

- To inspect a colony, you must open it up and look inside. Once inside, pry the outside frame of the brood chamber loose.
- Remove the frame from the body and hold it in front of you with one hand on each end of the top bar.
- If possible, position yourself so that the sun is shining over your shoulder and onto the frame.
- Observe the bees and the brood frame

Inspect the brood frames for

- **Healthy larvae**. Larvae should be pearly white. Gary, yellow, brown or black larvae are diseased, chilled or injured.

- **Eggs** standing in the bottom of cells. Recently laid eggs will be standing on end in the bottom of cells, one egg per cell. As they age, they gradually fall to one side. Two or more eggs on the sides of the cell are from a laying worker.

- **Cell caps** of healthy brood. These will be convex and tan. Cell caps of unhealthy brood are often concave and perforated with small holes.

- **Area of cells with brood**. A prolific queen will have laying pattern of brood with very few skipped cells over most of the frame. The pattern should be compact and in a semicircle, usually occurring over the bottom half of the frame.
- **Honey and pollen stores.** Honey should appear adjacent to the brood pattern. Adequate honey stores will vary with colony size. Pollen is stored in cells adjacent to honey.
- **Remove and inspect all of the frames that contain brood**. After inspection of a frame, place it in the hive body toward the side from which you removed the outside frame. After completing your inspection, replace the frames in their original order and close the hive.

Methods to Remove Bees from Super to Extract Honey

Removing a small amount of honey

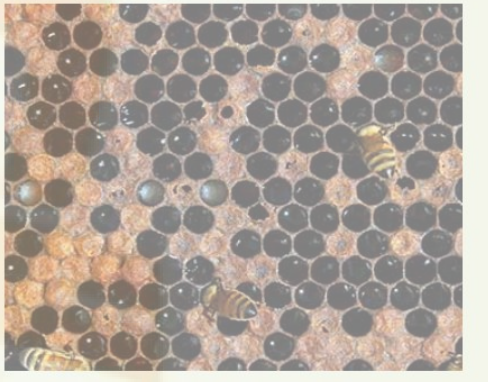
- Tapping of hive

Removing Surplus Honey

- Bee Repellents
- Air Blowers
- Bee Escape

Diseases of the Bees:

Bees suffer from different contagious diseases.



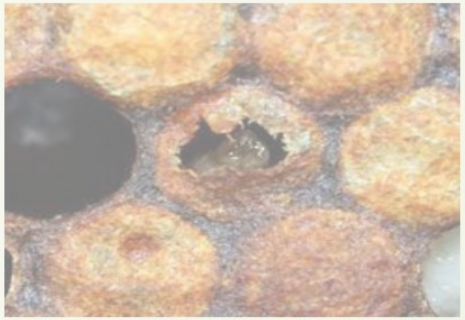
- **American foulbrood** – An extremely contagious disease of bees that affects them in the larval (worm) stage of development caused by the bacteria Bacillus larvae.

- **Deformed wing virus (DWV)** – an extremely common virus often associated with varroa mites that can also transmit the virus. Bees may show no symptoms or may have deformed wings, part of the symptoms of parasitic mite syndrome.

- **Dysentery** – A disease of honey bees causing an accumulation of excess waste products that are released in and near the hive.

- **Varroa mites** – Varroa destructor is considered the most serious cause of colony winter losses.

- **Sacbrood** – A slightly contagious disease of brood that is caused by sacbrood virus. Often associated with varroa mites



- **European foulbrood** – An infectious disease affecting honey bees in the larval (worm) stage of development; caused by the bacteria Streptococcus pluton. Extracted honey – Liquid honey

- **Nosema** – An infectious disease of the adult honey bee that infects the mid-gut, or stomach. It is cause by a protozoan parasite. Symptoms of this disease closely resemble those of dysentery

- **Tracheal mites** – Acarapiswoodi are microscopic and infest the breathing tubes inside the bees' thorax. Most bees are resistant to this parasite